

Lab 3: Sea-Level Rise Scenarios and Uncertainty

CEVE 421/521

2026-01-30

1 Overview

Sea-level rise projections vary widely depending on future emissions and our understanding of ice sheet dynamics. This lab explores two different approaches to characterizing this uncertainty:

1. **NOAA Scenarios:** Deterministic scenarios spanning a range of plausible futures (Low to High)
2. **BRICK Model:** Probabilistic projections with ensemble members representing parametric uncertainty

Learning Objectives:

1. Work with NetCDF files using YAXArrays (indexing, slicing, computing statistics)
2. Understand the difference between scenario-based and probabilistic projections
3. Compare sea-level rise projections across different emissions pathways (RCPs)
4. Interpret uncertainty ranges and how they grow with planning horizon
5. Apply these skills to real climate projection data for decision-making

2 Before Lab

Complete these steps **BEFORE** coming to lab:

1. **Accept the GitHub Classroom assignment** (link on Canvas)
 - This creates your personal copy of the lab repository
2. **Clone your repository:**

```
git clone https://github.com/CEVE-421-521/lab-03-S26-yourusername.git
cd lab-03-S26-yourusername
```

3. **Install and precompile packages** (this takes 5-10 minutes):

```
julia --project=. -e 'using Pkg; Pkg.instantiate()'
```

This downloads all required packages and compiles them. **Do this before lab** so you're ready to start working immediately.

4. **Open the notebook:**
 - In VS Code: Open the folder, then open `index.qmd`
 - Or use Quarto preview: `quarto preview index.qmd`

5. **Verify setup:** Run the first code cell under “Package Setup” to ensure packages load without errors.

⚠ Warning

Troubleshooting:

- If `Pkg.instantiate()` fails, check your internet connection
- If packages won't load, try restarting Julia
- If you get “package not found” errors, make sure you're in the lab directory when running Julia

3 Setup

3.1 Why Sewells Point, Norfolk, VA?

Norfolk, Virginia is one of the most vulnerable cities to sea-level rise in the United States. Sewells Point is a tide gauge location at the world's largest naval base (Naval Station Norfolk).

Why is Norfolk especially vulnerable?

- **Land subsidence:** The land is sinking due to groundwater extraction and post-glacial adjustment
- **Low elevation:** Much of the city is only a few feet above current sea level
- **Military significance:** Naval Station Norfolk is the world's largest naval base
- **Growing flood frequency:** The city already experiences regular “sunny day flooding”

All sea-level rise projections in this lab are for Sewells Point, measured in meters above the year 2000 baseline.

3.2 Package Setup

```
using Pkg
try
    # Activate the local project
    Pkg.activate(@__DIR__)

    # Check if Manifest exists, if not instantiate
    if !isfile(joinpath(@__DIR__, "Manifest.toml"))
        Pkg.instantiate()
    else
        # Try to resolve any inconsistencies
        Pkg.resolve()
    end
catch e
    @warn "Package environment setup failed, attempting recovery" exception=e
    Pkg.activate(@__DIR__)
    Pkg.instantiate()
end
```

```

using CairoMakie
using YAXArrays
using DimensionalData: dims
using NetCDF
using Statistics

include("sea_level_data.jl")

```

Line 1

For creating figures

Line 2

For working with multidimensional labeled arrays

Line 3

For working with dimensions

Line 4

For reading NetCDF data files

Line 5

For computing quantiles and means

Line 7

Helper functions for convenient data access (we'll build up to these)

4 Working with Climate Data

4.1 What is NetCDF?

NetCDF (Network Common Data Form) is the standard file format for climate and earth science data. It stores multidimensional arrays with labeled dimensions and metadata.

Think of NetCDF as a container that holds:

- **Data arrays** (e.g., sea level rise values)
- **Dimensions** (e.g., time, ensemble members, scenarios)
- **Coordinates** (e.g., specific years, RCP names)
- **Metadata** (e.g., units, descriptions, data sources)

4.2 Loading a NetCDF File

Let's open our sea-level data file and explore its structure:

```

data_path = joinpath(@__DIR__, "data", "sea_level_projections.nc")
ds = open_dataset(data_path)

```

YAXArray Dataset

Shared Axes:

(↓ time Sampled{Int64} 2000:1:2100 ForwardOrdered Regular Points)

```

Variables with additional axes:
  Additional Axes:
    (↓ noaa_scenario Sampled{Int64} 1:5 ForwardOrdered Regular Points)
  Variables:
    noaa_slr

  Additional Axes:
    (↓ ensemble Sampled{Int64} 1:1:1000 ForwardOrdered Regular Points,
    → rcp Sampled{Int64} 1:4 ForwardOrdered Regular Points)
  Variables:
    brick_slr

Properties: Dict{String, Any}("history" => "Generated by generate_sample_data.jl",
"note" => "For instructional purposes only", "source" => "Synthetic data based on
BRICK and NOAA patterns", "title" => "Sea Level Rise Projections for Sewells Point,
VA", "institution" => "Rice University CEVE 421/521")

```

The dataset contains multiple variables. Let's see what's inside:

```
ds
```

```

YAXArray Dataset
Shared Axes:
  (↓ time Sampled{Int64} 2000:1:2100 ForwardOrdered Regular Points)

Variables with additional axes:
  Additional Axes:
    (↓ noaa_scenario Sampled{Int64} 1:5 ForwardOrdered Regular Points)
  Variables:
    noaa_slr

  Additional Axes:
    (↓ ensemble Sampled{Int64} 1:1:1000 ForwardOrdered Regular Points,
    → rcp Sampled{Int64} 1:4 ForwardOrdered Regular Points)
  Variables:
    brick_slr

Properties: Dict{String, Any}("history" => "Generated by generate_sample_data.jl",
"note" => "For instructional purposes only", "source" => "Synthetic data based on
BRICK and NOAA patterns", "title" => "Sea Level Rise Projections for Sewells Point,
VA", "institution" => "Rice University CEVE 421/521")

```

You should see two main data variables:

- brick_slr: BRICK model projections (3D: time × ensemble × RCP)
- noaa_slr: NOAA scenarios (2D: time × scenario)

4.3 Exploring Dimensions and Coordinates

Let's examine the dimensions:

```
# What are the time points?  
ds["time"][:]
```

```
time 2000:1:2100
```

```
# How many ensemble members does BRICK have?  
ds["ensemble"][:]
```

```
ensemble 1:1:1000
```

💡 Your Response

Look at the output above.

What years does the data cover? How many ensemble members are in the BRICK projections?

Replace this text with your answer.

4.4 Indexing and Slicing

YAXArrays lets us select data by dimension names. Let's extract NOAA projections:

```
noaa_slr = ds["noaa_slr"]
```

```
┌ 101×5 YAXArray{Float32, 2} NOAA sea level rise scenario ┐  
└──────────────────────────────────────────────────────────┘  dims ┐  
  ↓ time Sampled{Int64} 2000:1:2100 ForwardOrdered Regular Points,  
  → noaa_scenario Sampled{Int64} 1:5 ForwardOrdered Regular Points  
└──────────────────────────────────────────────────────────┘  metadata ┐  
Dict{String, Any} with 7 entries:  
"units"    => "meters"  
"name"     => "noaa_slr"  
"location" => "Sewells Point, Norfolk, VA"  
"note"     => "Synthetic data for instructional purposes"  
"long_name" => "NOAA sea level rise scenario"  
"source"   => "Based on Sweet et al. 2017 patterns"  
"baseline" => "2000"  
└──────────────────────────────────────────────────────────┘  loaded lazily ┐  
data size: 1.97 KB  
└──────────────────────────────────────────────────────────┘
```

This is a 2D array with dimensions (time, scenario). Let's look at its size and dimensions:

```
size(noaa_slr)
```

```
(101, 5)
```

```
dims(noaa_slr)
```

```
(↓ time Sampled{Int64} 2000:1:2100 ForwardOrdered Regular Points,  
→ noaa_scenario Sampled{Int64} 1:5 ForwardOrdered Regular Points)
```

Now let's extract the "intermediate" scenario (index 3):

```
# Your code here
```

4.5 Computing Statistics

Let's work with the BRICK data, which has an ensemble dimension. First, extract the RCP 8.5 scenario (index 4):

```
brick_slr = ds["brick_slr"]  
rcp85_slr = brick_slr[rcp=4] # 4th RCP is RCP 8.5
```

```
┌ 101×1000 YAXArray{Float32, 2} BRICK local sea level rise projection ┐  
└───────────────────────────────────────────────────────────────────┘ dims ┌  
  ↓ time Sampled{Int64} 2000:1:2100 ForwardOrdered Regular Points,  
  → ensemble Sampled{Int64} 1:1:1000 ForwardOrdered Regular Points  
└───────────────────────────────────────────────────────────────────┘ metadata ┌  
Dict{String, Any} with 6 entries:  
"units"      => "meters"  
"name"       => "brick_slr"  
"location"   => "Sewells Point, Norfolk, VA"  
"note"       => "Synthetic data for instructional purposes"  
"long_name"  => "BRICK local sea level rise projection"  
"baseline"   => "2000"  
└───────────────────────────────────────────────────────────────────┘ loaded lazily ┌  
data size: 394.53 KB  
└───────────────────────────────────────────────────────────────────┘
```

This gives us a 2D array (time × ensemble). Let's compute the median across ensemble members for each year:

```
# Your code here
```

💡 Exercise

Compute the 95th percentile of sea-level rise across ensemble members for RCP 8.5.

Hint: Use `mapslices` with a function that computes the 95th percentile. You can use an anonymous function: `x -> quantile(x, 0.95)` Don't forget to use `dropdims` to remove the singleton ensemble dimension!

```
# Your code here
# Your code here
```

4.6 Quick Visualization

Now let's visualize what we just computed. We'll use the helper functions to make plotting easier:

```
let
  # Load data using helper functions (easier than raw YAXArrays for plotting)
  noaa_data = load_noaa_scenarios()
  brick_data = load_brick_projections()

  # Get specific scenarios
  noaa_int = get_noaa_scenario(noaa_data, "intermediate")
  brick_rcp85 = get_brick_scenario(brick_data, "rcp85")
  quantiles = brick_quantiles(brick_rcp85, [0.50, 0.95])

  fig = Figure(size=(700, 400))
  ax = Axis(
    fig[1, 1];
    xlabel="Year",
    ylabel="Sea Level Rise (meters)",
    title="Sea-Level Rise: First Look"
  )

  # NOAA intermediate scenario
  lines!(ax, noaa_data.years, noaa_int; label="NOAA Intermediate", color=:blue,
  linewidth=2)

  # BRICK RCP 8.5 median and 95th percentile
  lines!(ax, brick_data.years, quantiles[:, 1]; label="BRICK RCP 8.5 Median",
  color=:red, linewidth=2)
  lines!(ax, brick_data.years, quantiles[:, 2]; label="BRICK RCP 8.5 (95th
  percentile)",
    color=:red, linewidth=2, linestyle=:dash)

  axislegend(ax; position=:lt)
  fig
end
```

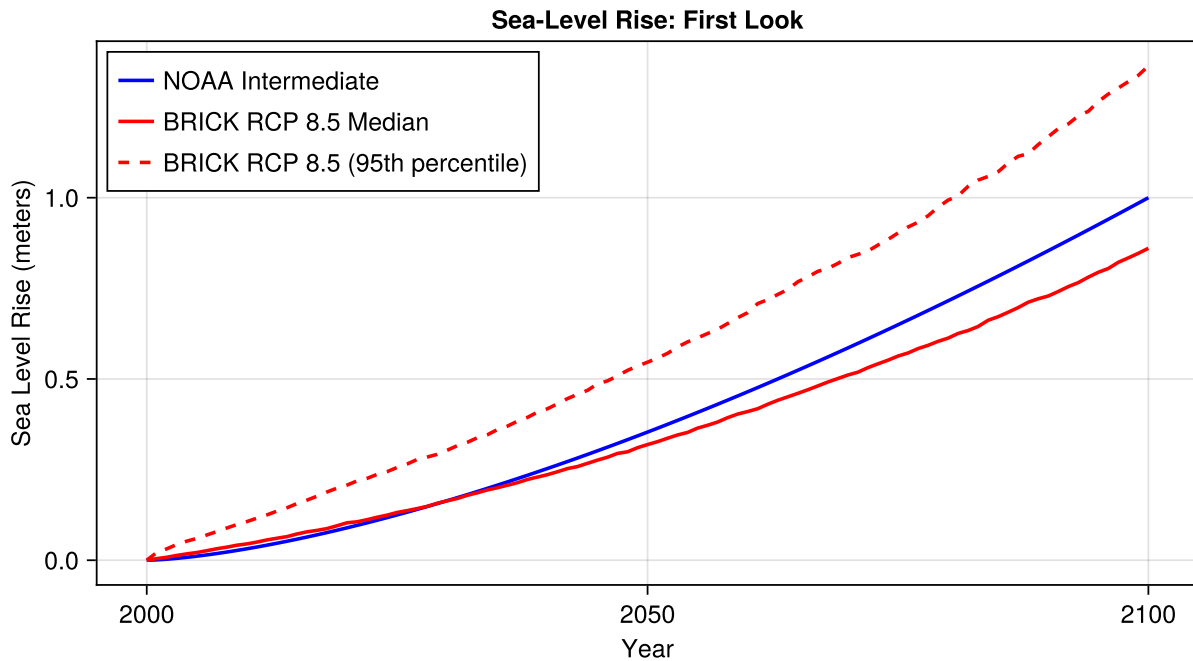


Figure 1: First look at sea-level rise projections: comparing NOAA and BRICK data.

💡 Your Response

What do you notice about the relationship between the NOAA Intermediate scenario and the BRICK RCP 8.5 projections?

Replace this text with your observations.

5 NOAA Sea-Level Rise Scenarios

NOAA provides deterministic scenarios that span a range of plausible futures. These are NOT probabilistic projections — they don't have probabilities attached. Instead, they represent "what if" storylines.

Now that you've learned the basics of working with NetCDF and YAXArrays, let's use some helper functions to make our analysis more convenient.

5.1 Loading the Data

```
noaa_data = load_noaa_scenarios()
println("NOAA Scenarios: $(noaa_data.scenarios)")
println("Years: $(noaa_data.years[1]) to $(noaa_data.years[end])")
```

```
NOAA Scenarios: ["low", "int_low", "intermediate", "int_high", "high"]
Years: 2000 to 2100
```


5.2 Visualizing the Scenarios

```
let
  fig = Figure(size=(700, 450))
  ax = Axis(
    fig[1, 1];
    xlabel="Year",
    ylabel="Sea Level Rise (meters)",
    title="NOAA Sea-Level Rise Scenarios for Sewells Point, VA"
  )

  colors = [:green, :blue, :gold, :orange, :red]
  labels = ["Low", "Intermediate-Low", "Intermediate", "Intermediate-High", "High"]

  for (i, scenario) in enumerate(noaa_data.scenarios)
    slr = get_noaa_scenario(noaa_data, scenario)
    lines!(ax, noaa_data.years, slr; color=colors[i], linewidth=2,
label=labels[i])
  end

  axislegend(ax; position=:lt)
  fig
end
```

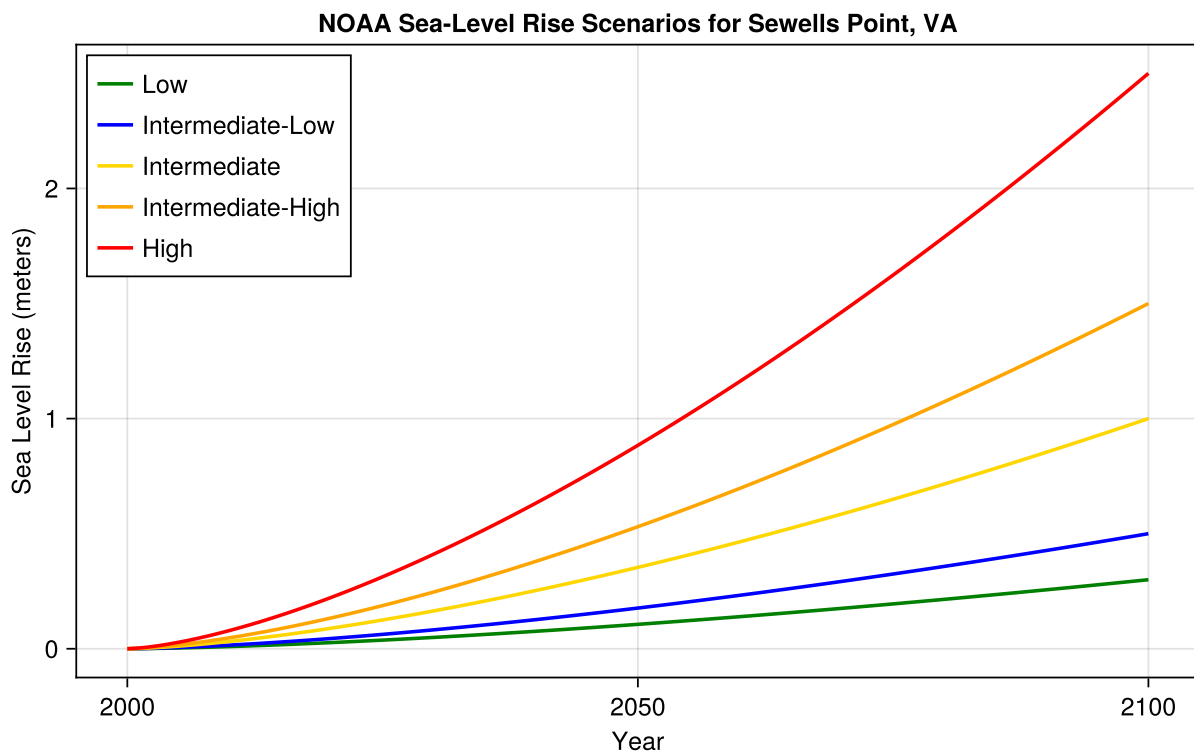


Figure 2: NOAA sea-level rise scenarios for Sewells Point, Norfolk, VA. Each scenario represents a plausible future, not a probability.

5.3 Interpreting the Scenarios

💡 Your Response

Look at Figure 2. By 2100, the “High” scenario projects about 2.5 meters of sea-level rise, while “Low” projects only 0.3 meters.

If you were advising a city planner, how would you explain what these scenarios mean? Why isn’t there a probability attached to each scenario?

Replace this text with your answer.

6 BRICK Model Projections

The BRICK model (Building blocks for Relevant Ice and Climate Knowledge) takes a different approach. For each emissions pathway (RCP), it generates an **ensemble** of projections representing parametric uncertainty.

6.1 Loading BRICK Data

```
brick_data = load_brick_projections()
println("RCP Scenarios: ${brick_data.rcps}")
println("Ensemble members: ${length(brick_data.ensemble)}")
println("Years: ${brick_data.years[1]} to ${brick_data.years[end]}")
```

```
RCP Scenarios: ["rcp26", "rcp45", "rcp60", "rcp85"]
Ensemble members: 1000
Years: 2000 to 2100
```

6.2 Visualizing a Single RCP

Let’s look at RCP 8.5 (high emissions) first:

```
let
    rcp85 = get_brick_scenario(brick_data, "rcp85")
    quantiles = brick_quantiles(rcp85, [0.05, 0.50, 0.95])

    fig = Figure(size=(700, 450))
    ax = Axis(
        fig[1, 1];
        xlabel="Year",
        ylabel="Sea Level Rise (meters)",
        title="BRICK RCP 8.5 Projections (${length(brick_data.ensemble)} ensemble
members)"
    )

    # Plot a sample of ensemble members
    for i in 1:50:length(brick_data.ensemble)
        lines!(ax, brick_data.years, rcp85[:, i]; color=(:gray, 0.2), linewidth=0.5)
    end
```

```

# Plot quantiles
lines!(ax, brick_data.years, quantiles[:, 1]; color=:blue, linewidth=2,
linestyle=:dash, label="5th percentile")
lines!(ax, brick_data.years, quantiles[:, 2]; color=:red, linewidth=2,
label="Median")
lines!(ax, brick_data.years, quantiles[:, 3]; color=:blue, linewidth=2,
linestyle=:dash, label="95th percentile")

axislegend(ax; position=:lt)
fig
end

```

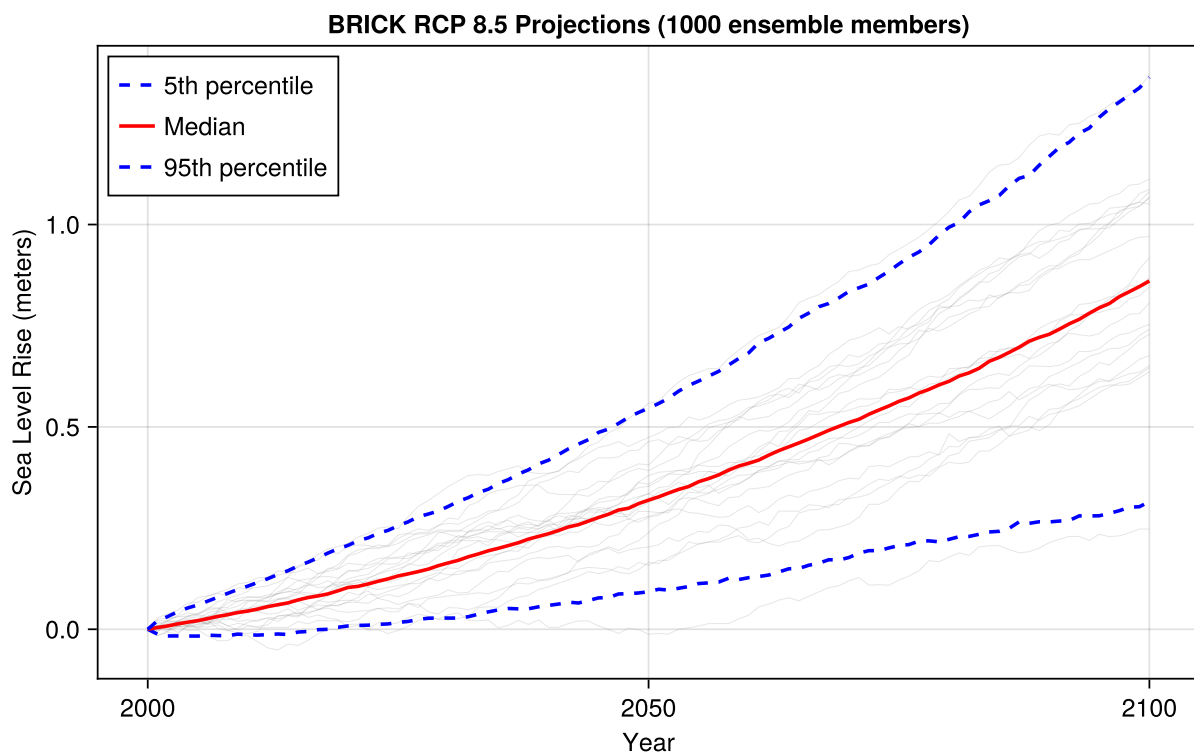


Figure 3: BRICK sea-level rise projections for RCP 8.5 at Sewells Point. Gray lines show individual ensemble members; colored lines show quantiles.

6.3 Comparing RCP Scenarios

Now let's compare projections across all four RCP scenarios:

```

let
  fig = Figure(size=(700, 450))
  ax = Axis(
    fig[1, 1];
    xlabel="Year",
    ylabel="Sea Level Rise (meters)",

```

```

        title="BRICK Projections by RCP Scenario"
    )

    colors = Dict("rcp26" => :green, "rcp45" => :blue, "rcp60" => :orange, "rcp85"
=> :red)
    labels = Dict("rcp26" => "RCP 2.6", "rcp45" => "RCP 4.5", "rcp60" => "RCP 6.0",
"rcp85" => "RCP 8.5")

    for rcp in brick_data.rcps
        scenario_data = get_brick_scenario(brick_data, rcp)
        quantiles = brick_quantiles(scenario_data, [0.05, 0.50, 0.95])

        # Shaded uncertainty range
        band!(ax, brick_data.years, quantiles[:, 1], quantiles[:, 3];
            color=(colors[rcp], 0.2))

        # Median line
        lines!(ax, brick_data.years, quantiles[:, 2];
            color=colors[rcp], linewidth=2, label=labels[rcp])
    end

    axislegend(ax; position=:lt)
    fig
end

```

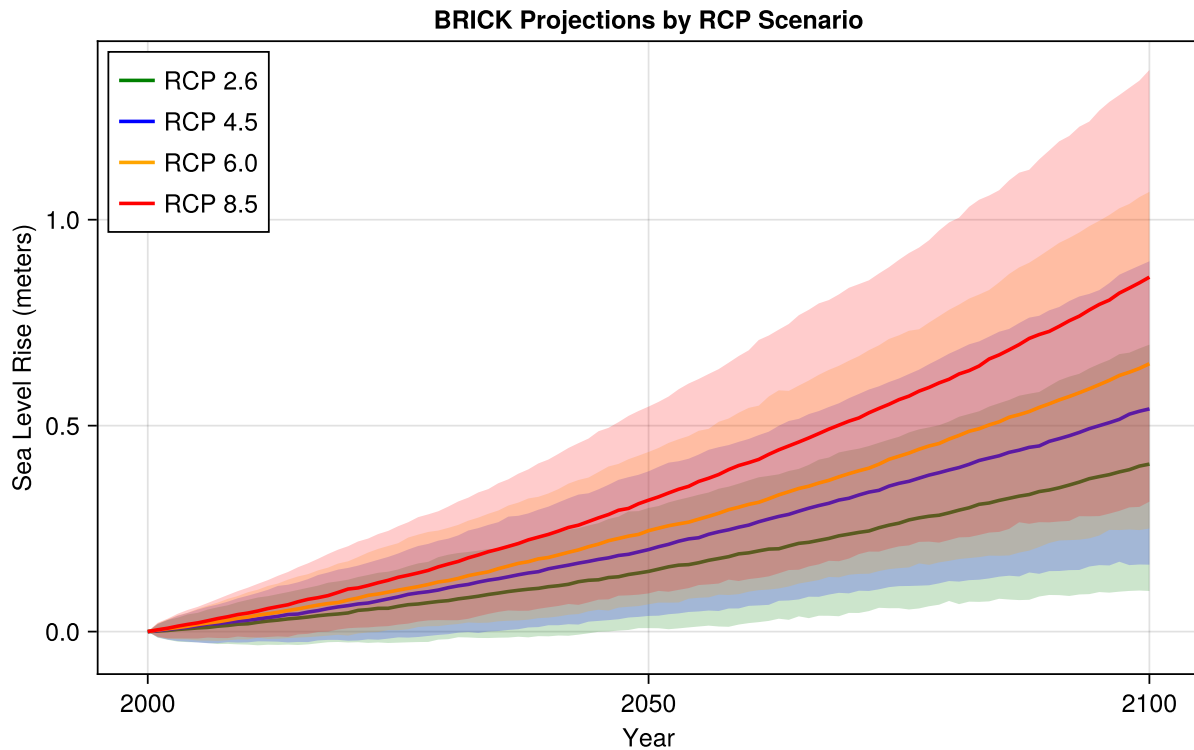


Figure 4: BRICK projections across all RCP scenarios. Shaded regions show 5th-95th percentile ranges; lines show medians.

6.4 Understanding the Spread

Let's look at the distribution of projections at 2050 and 2100 for RCP 8.5:

```
let
  slr_2050 = slr_at_year(brick_data, "rcp85", 2050)
  slr_2100 = slr_at_year(brick_data, "rcp85", 2100)

  fig = Figure(size=(700, 350))

  ax1 = Axis(fig[1, 1]; xlabel="Sea Level Rise (meters)", ylabel="Density",
title="2050")
  hist!(ax1, slr_2050; bins=30, normalization=:pdf, color=(red, 0.6))
  vlines!(ax1, [median(slr_2050)]; color=:black, linewidth=2, linestyle=:dash,
label="Median")

  ax2 = Axis(fig[1, 2]; xlabel="Sea Level Rise (meters)", ylabel="Density",
title="2100")
  hist!(ax2, slr_2100; bins=30, normalization=:pdf, color=(red, 0.6))
  vlines!(ax2, [median(slr_2100)]; color=:black, linewidth=2, linestyle=:dash)

  fig
end
```

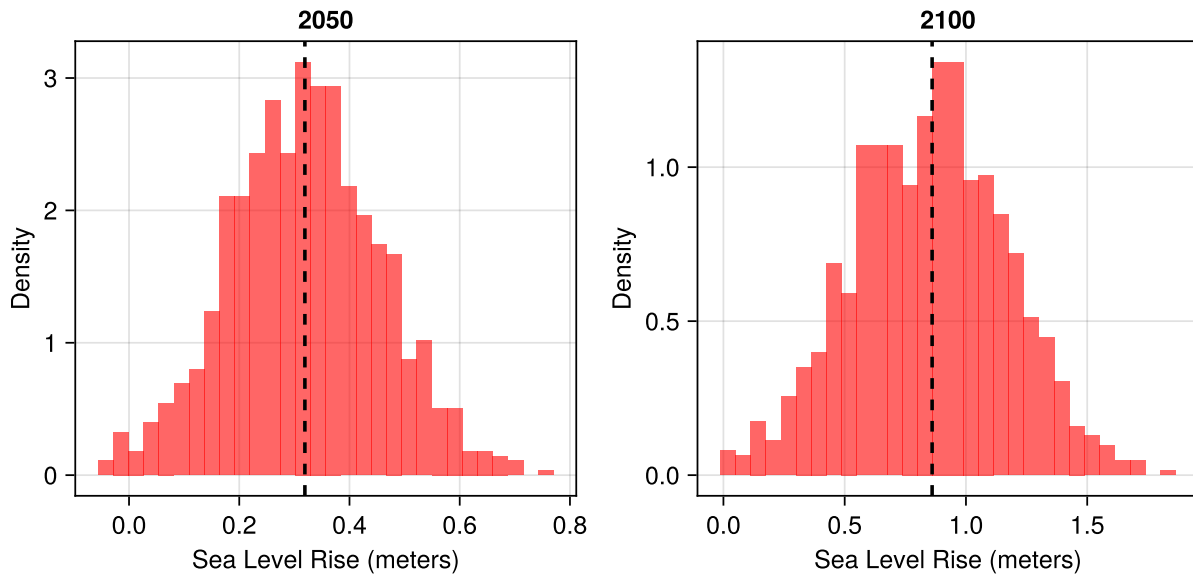


Figure 5: Distribution of sea-level rise projections at 2050 and 2100 for RCP 8.5.

💡 Your Response

Compare the distributions at 2050 and 2100 in Figure 5.

How does the spread (uncertainty) change as we project further into the future? Why does this happen?

Replace this text with your answer.

7 Comparing Approaches

Now let's overlay the NOAA scenarios and BRICK projections to see how they compare:

```
let
  rcp85 = get_brick_scenario(brick_data, "rcp85")
  quantiles = brick_quantiles(rcp85, [0.05, 0.50, 0.95])

  fig = Figure(size=(700, 500))
  ax = Axis(
    fig[1, 1];
    xlabel="Year",
    ylabel="Sea Level Rise (meters)",
    title="NOAA Scenarios vs BRICK RCP 8.5"
  )

  # BRICK uncertainty range
  band!(ax, brick_data.years, quantiles[:, 1], quantiles[:, 3];
    color=(red, 0.2), label="BRICK 90% range")
  lines!(ax, brick_data.years, quantiles[:, 2];
    color=red, linewidth=2, label="BRICK median")
```

```

# NOAA scenarios
noaa_colors = [:green, :blue, :gold, :orange, :purple]
noaa_labels = ["NOAA Low", "NOAA Int-Low", "NOAA Int", "NOAA Int-High", "NOAA
High"]

for (i, scenario) in enumerate(noaa_data.scenarios)
    slr = get_noaa_scenario(noaa_data, scenario)
    lines!(ax, noaa_data.years, slr;
           color=noaa_colors[i], linewidth=2, linestyle=:dash,
           label=noaa_labels[i])
end

axislegend(ax; position=:lt, nbanks=2)
fig
end

```

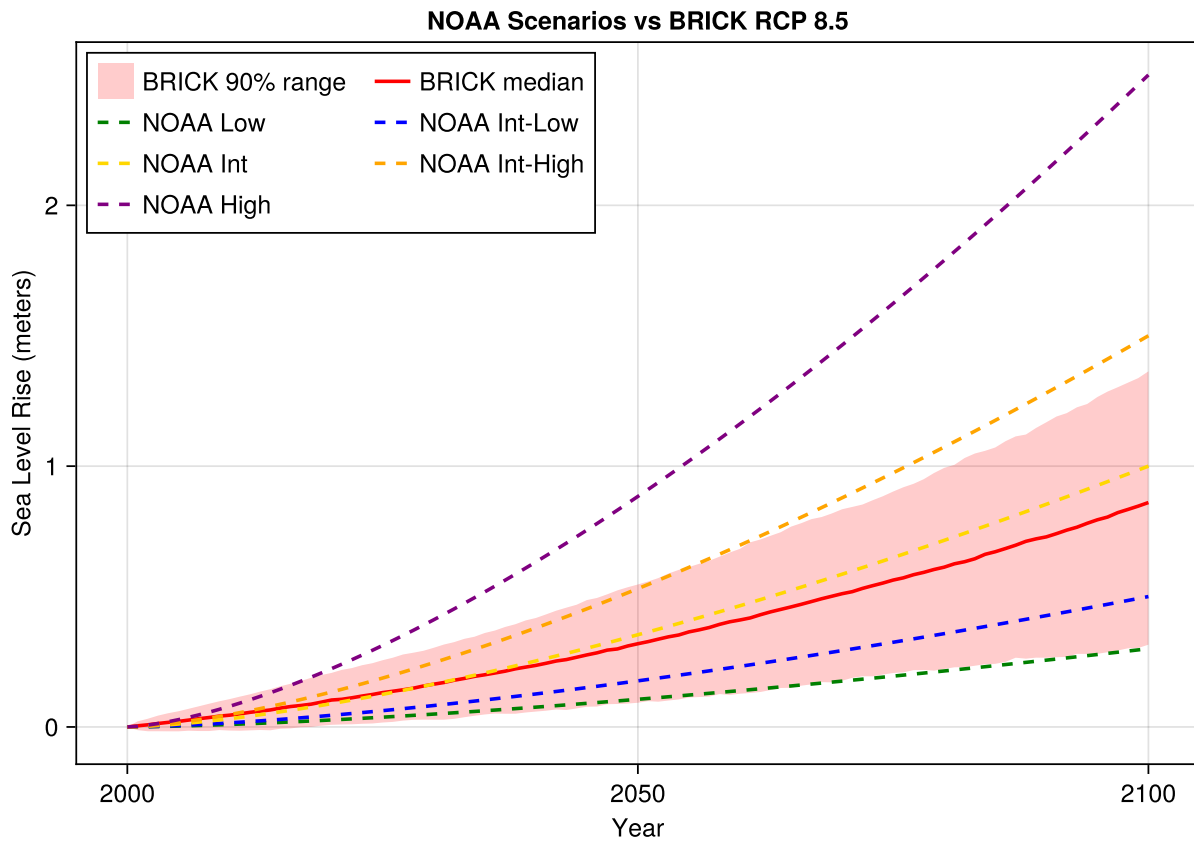


Figure 6: Comparison of NOAA scenarios (dashed lines) and BRICK RCP 8.5 projections (shaded region with solid median).

7.1 Key Observations

Your Response

Looking at Figure 6, answer these questions:

1. *Where does the NOAA “High” scenario fall relative to the BRICK 90% range?*
2. *What does this tell us about the relationship between these two approaches?*
3. *The BRICK ensemble spread represents uncertainty in model parameters. What kinds of uncertainties might this capture?*
4. *The choice between NOAA scenarios represents different possible futures. Why can’t we assign probabilities to these scenarios?*

Replace this text with your answers.

8 Reflection

8.1 Connecting to Lecture Concepts

Think back to this week’s lectures on climate scenarios and projections.

Question 1

In the context of sea-level rise projections:

- *What is the difference between a scenario (like NOAA’s “High” scenario) and a projection (like BRICK’s probabilistic ensemble)?*
- *Why do climate scientists use scenarios instead of making single predictions about the future?*

Replace this text with your answer.

8.2 Planning Horizons

Question 2

Consider two planning decisions:

1. *Designing a seawall that must last until 2050*
2. *Zoning restrictions for development in flood-prone areas (relevant through 2100 and beyond)*

How would you use these projections differently for each decision? Which approach (NOAA scenarios or BRICK probabilistic) would you prefer for each, and why?

Replace this text with your answer.

8.3 Synthesis

💡 Question 3

A city council member says: “These scientists can’t even agree on how much sea level will rise. Why should we spend money on adaptation when they don’t know what will happen?”

How would you respond? Use concepts from both the lecture and this lab.

Replace this text with your answer.

9 Summary

In this lab, you learned to:

1. **Work with NetCDF files** using YAXArrays (indexing, slicing, computing statistics across dimensions)
2. **Distinguish** between scenario-based (NOAA) and probabilistic (BRICK) approaches to projections
3. **Compare** projections across emissions pathways (RCPs)
4. **Interpret** uncertainty ranges and how they grow with planning horizon
5. **Apply** these data skills to real climate projection data for decision-making

The key insight: **Uncertainty is not a reason to delay action.** Both scenarios and probabilistic projections are tools for making better decisions under uncertainty.

10 Submission

1. Push your completed notebook to GitHub
2. If the formatter creates a pull request, approve it and re-render
3. Submit the link to your repository on Canvas

Checklist

Before submitting, verify you have completed:

- ☐ **Setup:** All packages load without error
- ☐ **NetCDF basics:** Answered questions about dimensions and coordinates
- ☐ **YAXArrays exercise:** Computed 95th percentile for RCP 8.5
- ☐ **First visualization:** Described observations from initial plot
- ☐ **NOAA interpretation:** Explained what scenarios mean vs probabilities
- ☐ **BRICK distributions:** Explained why uncertainty grows with time
- ☐ **Comparison:** Answered all four comparison questions
- ☐ **Reflection Q1:** Explained scenarios vs projections
- ☐ **Reflection Q2:** Discussed different uses for different planning horizons
- ☐ **Reflection Q3:** Addressed the skeptical council member
- ☐ **Render:** Re-rendered the notebook to ensure all outputs are current

11 References

Bibliography